Single-atom detection versus camera parameters at 852 nm

P1 interim technical report: validation, camera presets, sensitivity and Sobol analyses

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This report covers the shared package `camsim_common` and project P1 through the validation stage, the camera presets and sensitivity study with the measured V1 point-spread function (PSF), and two interim Sobol analyses. It supersedes the summary sections of `reports/p1_production_note.md`, which uses a diffraction-limited PSF. Detailed results are in `reports/p1_validation.md`, `reports/p1_sobol_interim_n256.md` and `reports/p1_sobol_camera_n256.md`. All results are simulations (SIM) unless marked otherwise, and every SIM value should be reviewed before submission.

Summary

We built a semi-analytic model that maps a camera description (`SensorModel`) and the photon budget and loss of a trapped Cs atom to the false-positive (FP) and false-negative (FN) rates of single-site occupancy detection, and from them to the imaging time needed for a target infidelity ε. The model reproduces the FN rate of Manetsch *et al.* [1] to within 11% and the readout statistics of Picken *et al.* [2], but under-predicts the FP rate of Manetsch *et al.* by a factor of 14, which we attribute to events outside a camera model.

In a reference scenario anchored on Manetsch *et al.* (NA 0.55, measured PSF of FWHM 1.70 µm, 2.7% of scattered photons reaching the camera), five low-noise CMOS configurations reach $\varepsilon = 10^{-4}$ in 40 to 44 ms with a matched-filter classifier, and an EMCCD in multi-frame photon counting in 34 ms. The Prime BSI Express in CMS mode needs 28.4 ms for 10^{-3} and 41.3 ms for 10^{-4}, within 20% of the best camera in the set. Cameras with a quantum efficiency (QE) at 852 nm below 0.25 need 2.5 to 4 times longer than the CMOS group or do not reach 10^{-4}.

Over the camera-side parameter space, the imaging time is set by QE and read noise, which act separably: at the reference background and 1 pixel per PSF FWHM, t is proportional to 1/QE times a read-noise factor that is 0.77 at 0.1 e^-, 0.84 at 0.5 e^-, 1 at 1.0 e^- and 1.42 at 2.0 e^- (for 10^{-4}). Hence QE is the main lever for a Stage 1 sensor, and a read-noise median below 0.5 e^- gains less than 10%. Two inputs that carry little variance globally matter at the operating point: a 2% fraction of pixels at five times the median read noise slows a 1 e^- camera by 42%, and the loss per scattered photon decides whether 10^{-4} is reachable at all. NA and atom loss, which are properties of the experiment, outweigh every camera parameter.

The main limitations are (i) the unexplained FP floor of the V1 experiment, (ii) an assumed two-class read-noise tail in place of a measured distribution, and (iii) a single reference scenario at 852 nm; Rb at 780 nm and array crosstalk are not yet treated.

1 Scope and what was run

The P1 brief asks three questions: (i) which sensor parameters set the infidelity and imaging time, and with what sensitivity; (ii) what minimum specification reaches 10^{-3} and 10^{-4} within a given imaging time and loss budget (the Stage 1 camera specification); and (iii) how the lab should operate the Prime BSI Express. This report gives interim answers to (i) and (iii) and the elements of (ii).

We ran (a) validation stages V1 to V3 with a convergence study; (b) eleven camera presets at seven pixel samplings, with the diffraction-limited and with the measured PSF; (c) a sensitivity

study of the preset results to their uncertain inputs; (d) a reduced generic CMOS-class sweep of 750 points with the diffraction-limited PSF; (e) Sobol analyses at $N = 256$ of an 8-parameter design (camera parameters, NA and loss) and of a 6-parameter camera-only design (NA and loss fixed). Table 7 lists the cost of each.

2 Model and validation

Here, an atom present for the full exposure t yields

$$\mu = R\, t\, \eta_{\mathrm{cam}}\, \mathrm{QE} \tag{1}$$

detected photoelectrons, with R the scattering rate, $\eta_{\mathrm{cam}} = \eta_{\mathrm{coll}}(\mathrm{NA})\, \eta_{\mathrm{opt}}$ the fraction of scattered photons reaching the sensor and QE the quantum efficiency at 852 nm. Loss during imaging truncates the exposure; model A uses a constant loss probability per scattered photon, model B a 3D Langevin walk in the trap. Per-pixel counts are Poisson in signal plus background, convolved with a Gaussian-mixture read-noise model (core plus tail pixels) or, for an EMCCD, a gamma-distributed multiplication register with clock-induced charge (CIC) [3]. Site classification uses a box sum, a matched filter with Fisher-optimal pixel weights, or a per-pixel Bayesian likelihood ratio. Count distributions are computed exactly on a lattice or, for linear statistics, from characteristic functions (within 2% of the lattice result in optimal infidelity, about 10^3 times faster), and cross-checked against brute-force and importance-sampled Monte Carlo at 10^{-3}. Thresholds are optimised per site and averaged over 3×3 sub-pixel site positions (within 1.1% of a 9×9 average).

Table 1: Validation results. Details and residuals in `reports/p1_validation.md`.

stage	target (source)	model result	status
V1	FN 5.50×10^{-5}, FP 7.01×10^{-5}, ORCA-Quest, 80 ms [1]	FN 4.91×10^{-5} (empirical rate spread) to 5.73×10^{-5} (Gaussian); FP 4.85×10^{-6}	FN reproduced; FP 14× low
V1	dependence on loss-time profile at fixed survival 0.999864	FN −42% (model B, late losses) to +10% (model B, early escape) relative to model A	quantified
V2	Zyla 5.5, 40 ms, readout error $< 8 \times 10^{-7}$ [2]	fitted gains 0.381 and 0.368 e^-/ADU (agree between traps); Poisson readout error 2.9 to 4.0 times their Gaussian estimate (9.55×10^{-7} for trap 1)	reproduced; bound marginal
V2	sCMOS vs EMCCD SNR table (their Table I)	all cells to 0.05 except two EMCCD 852 nm cells, consistent with QE 0.55 instead of 0.60	reproduced
V3	Poisson limit; EMCCD excess noise $F^2 \to 2$ [3]	relative error 4.42×10^{-14}; $F^2 = 1.972$ at gain 1000, 512 stages	pass

The V1 FP excess is visible directly in the digitised histogram: between 32 and 42 photons the measured background tail lies 2 to 3 times above any Poisson-plus-read-noise model consistent with the background width. We read this as a floor set by events outside the camera, for example atoms entering during imaging. This has a direct consequence for the present work: at the 10^{-4} level, a real experiment may be limited by non-camera errors that no camera choice removes.

3 Reference scenario

The reference scenario (`configs/production/reference_cs.yaml`) is derived from the V1 fit: Cs D2 at 852 nm [4]; scattering rate 72.3×10^3/s; NA 0.55 with $\eta_{\mathrm{opt}} = 0.327$ (2.7% of scattered photons to the camera, as in V1); loss 2.35×10^{-8} per scattered photon (V1 survival, model A); site-to-site rate spread 12.3% rms (V1 Ext. Data Fig. 4d); background 4.76 e^-/px/s at the V1 pixel (0.902 µm in the object plane, QE 0.44), scaled with pixel area and QE; filling 0.5. The PSF is

a Gaussian of $1/e^2$ radius 1.44 µm at NA 0.55 (FWHM 1.70 µm), derived from the waist stated by Manetsch *et al.* with the pixel variance removed, and scaled as 1/NA when NA is varied. Its FWHM is 2.1 times that of the diffraction-limited vectorial PSF (0.82 µm). Survival is not limiting in this scenario: the loss rate is 1.7×10^{-3}/s, so survival stays above 0.99 up to 5.9 s (DERIVED: $0.01005/(2.35 \times 10^{-8} \times 72.3 \times 10^{3}/\mathrm{s})$).

4 Camera presets

Table 2: Presets at their best pixel sampling, measured PSF. Matched filter; photon counting uses an SNR-optimal box over frames at 670 fps (128×128 crop); HCG: high conversion gain. QE at 852 nm digitised from vendor typical curves (DATA, about ±1.5 to 2%). Magnification for the V1 PSF (FWHM 1.70 µm). n/r: not reached within 300 ms with survival ≥ 0.99. The last column gives $t(10^{-4})$ with the diffraction-limited vectorial PSF, each at its own best sampling.

camera	mode	QE 852 nm	read noise med./rms (e^-)	best px/FWHM	magni-fication	$t(10^{-3})$ (ms)	$t(10^{-4})$ (ms)	$t(10^{-4})$ diffr. (ms)
iXon Ultra 888 EXF	photon counting	0.608	EM ×1000	1.4	10.7	21.9	34.3	29.6
ORCA-Fusion BT	ultra quiet	0.464	0.60/0.70	1.0	3.8	26.7	39.7	36.9
ORCA-Quest 2	ultra quiet (PNR)	0.384	0.25/0.30	1.4	3.8	26.4	40.2	35.7
Prime BSI Express	CMS (100 MHz)	0.485	1.00/1.00	1.0	3.8	28.4	41.3	37.5
ORCA-Quest 2	standard	0.384	0.39/0.43	1.0	2.7	27.9	42.6	37.9
Marana 4.2B-6	low noise 12-bit	0.470	1.10/1.10	1.0	3.8	30.3	43.6	39.6
iXon Ultra 888 EXF	analog	0.608	EM ×1000	3.0	23.0	30.2	48.1	42.5
Prime BSI Express	Speed (200 MHz)	0.485	1.60/1.60	0.7	2.7	34.0	49.3	46.3
QHY268M (IMX571)	HG-2CMS, g. 100	0.208	1.19/1.19	0.7	1.6	68.8	104.2	94.4
Zyla 5.5	global shutter	0.249	2.30/2.50	0.5	1.9	94.6	156.2	154.2
Blackfly S (IMX425)	HCG	0.186	4.69/4.69	0.35	1.9	179.1	n/r	299.6

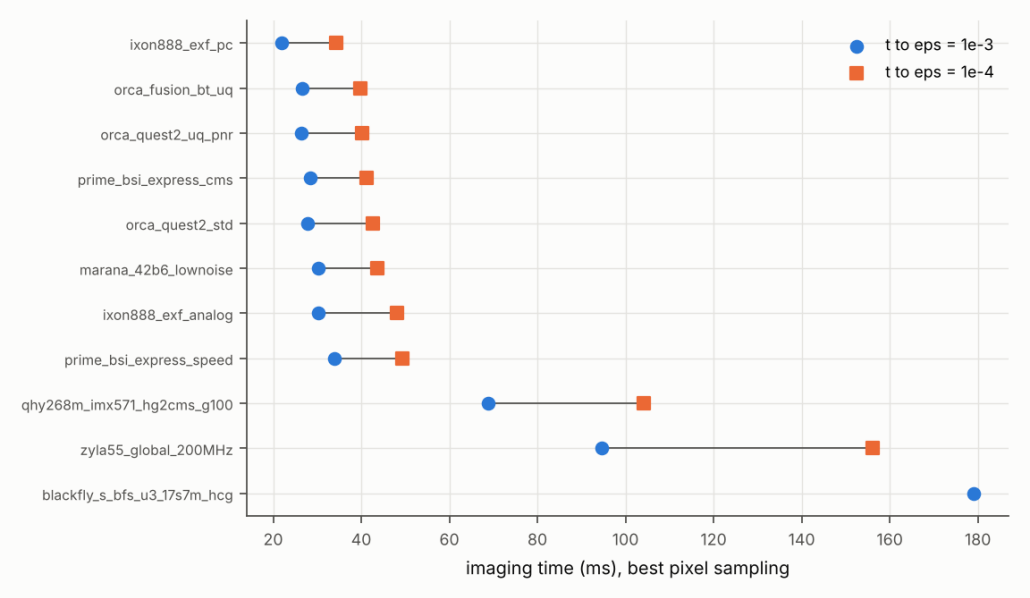


Figure 1: Imaging time to $\varepsilon = 10^{-3}$ (circles) and 10^{-4} (squares) for each preset at its best pixel sampling (0.35 to 3 px per FWHM), reference Cs scenario (Sec. 3) with the measured V1 PSF. Characteristic-function engine with per-site matched filter; EMCCD photon counting uses Poisson-binomial counts over an SNR-optimal box with frames at 670 fps. Error rates averaged over 3×3 sub-pixel site positions, threshold optimised per site, filling 0.5. The Blackfly S does not reach 10^{-4} within 300 ms. Data: `runs/production/prod_presets/prod_presets.h5`.

As can be seen from Table 2 and Fig. 1, the measured PSF, although 2.1 times wider than the diffraction limit, lengthens the imaging time by only 1 to 16%. This is because the best magnification

changes little (3.8 instead of 4.0 for the Prime BSI Express): the optimal sampling moves from about 0.5 to about 1 px per FWHM, and the wider PSF costs only the read noise and background of the additional pixels in the classifier window. The ranking is unchanged except that the ORCA-Fusion BT and the ORCA-Quest 2 in PNR mode exchange places, a difference of 1.3%. Three remarks. (i) The iXon advantage requires photon counting over many short frames (26 frames in 40 ms for a 128×128 crop); in analog mode the excess noise factor ($F^2 = 1.97$) cancels its QE advantage. (ii) With the QE of 0.44 stated by Manetsch *et al.* for their ORCA-Quest, instead of 0.384 from the Quest 2 typical curve, the Quest 2 times would fall by about 13% (t roughly proportional to 1/QE).[1] (iii) Differences below about 5% between presets are within the QE uncertainty and should not be used to rank them.

5 Sensitivity of the preset results

Table 3: Changes to the preset results, measured PSF. Prime BSI Express CMS at 1.0 px per FWHM unless stated. The high loss, 5.6×10^{-7} per photon, is the bound implied by the retention stated in Ref. [2].

change	$t(10^{-3})$ (ms)	$t(10^{-4})$ (ms)	change in $t(10^{-4})$
baseline	28.4	41.3	
QE 0.47 / 0.50 (digitisation range)	29.3 / 27.5	42.6 / 40.0	+3% / −3%
2% of pixels at 5× median read noise	37.9	58.7	+42%
loss 5.6×10^{-7} per photon (24× reference)	29.5	n/r	10^{-4} not reachable
ORCA-Quest 2 PNR, loss 5.6×10^{-7}	27.7	n/r	10^{-4} not reachable
iXon PC, CIC all in image area / all in register	23.7 / 23.1	37.7 / 36.0	5% between the limits

Hence the read-noise tail, which no vendor datasheet in the set specifies, is the most consequential camera input after QE and the median (Table 3), and atom loss can make 10^{-4} unreachable whatever the camera: at the high loss, ε at 40 ms is 3.6×10^{-4} for the Prime BSI Express and is set by atoms lost during the exposure.

6 Global sensitivity

The brief asks for Sobol indices of the time to reach 10^{-3} and 10^{-4} at survival ≥ 0.99. We ran two designs (Saltelli sampling in SALib [5], no second-order terms, 95% bootstrap confidence intervals), each as the first $N = 256$ base points of a nested $N = 1024$ design. The output is $\log_{10} t$, with targets not reached within the scan censored at the scan limit.

8-parameter design (2560 points; scan to 1 s). Over QE 0.2 to 0.95, read-noise median 0.1 to 4 e^-, tail fraction 0 to 5%, 0.35 to 2 px per FWHM, background 0.1 to 3 times the reference, dark current 10^{-3} to 3 e^-/px/s, NA 0.3 to 0.8 and loss 2×10^{-9} to 6×10^{-7} per photon, NA dominates $t(10^{-3})$ ($S_T = 0.56 \pm 0.13$), followed by read noise (0.22 ± 0.07) and QE (0.14 ± 0.04). At 10^{-4}, 31% of points are censored, mostly at high loss and low NA, so the indices mix the time to reach the target with whether it is reached; NA (0.51 ± 0.12), loss (0.34 ± 0.09) and read noise (0.30 ± 0.08) are not separable at this N (Fig. 2). Over NA 0.3 to 0.8, the collection fraction for isotropic emission, $\eta_{\mathrm{coll}} = (1 - \sqrt{1 - \mathrm{NA}^2})/2$, rises from 0.023 to 0.200 (DERIVED), a factor of 8.7, against a factor of 4.75 across the QE range.

Camera-only design (2048 points; NA 0.55 and loss fixed at the reference; scan to 5 s, 3% censored at 10^{-4}). Table 4 and Fig. 3 give the indices.

[1] The vendor's typical curve for the original ORCA-Quest (C15550-20UP) gives 0.396 at 852 nm, so the 0.44 of Ref. [1] may reflect a particular unit or a different definition; we have not resolved this.

Table 4: Camera-only Sobol indices of $\log_{10} t$ (S_1: first order; S_T: total; $\pm$ 95% bootstrap confidence interval), $N = 256$. Inputs sampled uniformly, or log-uniformly for read-noise median, sampling, background scale and dark current, over the ranges of the 8-parameter design.

parameter	S_1, $t(10^{-4})$	S_T, $t(10^{-4})$	S_1, $t(10^{-3})$	S_T, $t(10^{-3})$
read-noise median	0.60 ± 0.15	0.65 ± 0.12	0.54 ± 0.13	0.59 ± 0.09
QE	0.34 ± 0.10	0.38 ± 0.08	0.35 ± 0.09	0.34 ± 0.06
background scale	0.05 ± 0.05	0.06 ± 0.02	0.05 ± 0.05	0.06 ± 0.02
pixels per FWHM	0.00 ± 0.02	0.04 ± 0.02	0.00 ± 0.03	0.03 ± 0.01
read-noise tail fraction	0.02 ± 0.02	0.02 ± 0.02	0.03 ± 0.03	0.02 ± 0.01
dark current	0.00	0.00	0.00	0.00

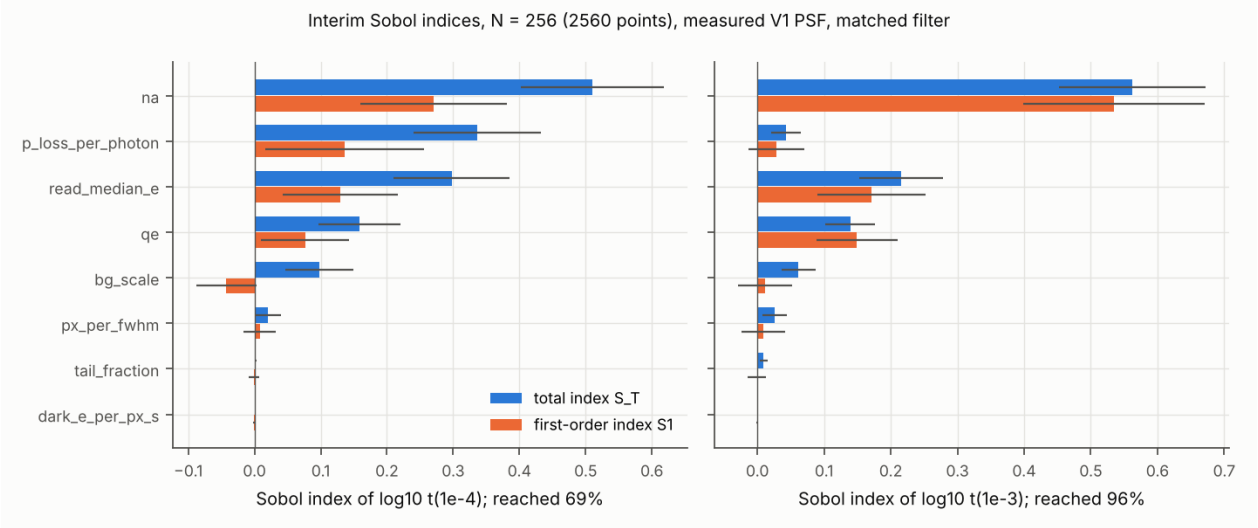


Figure 2: Sobol indices of $\log_{10} t(10^{-4})$ (left) and $\log_{10} t(10^{-3})$ (right) for the 8-parameter design, $N = 256$ (2560 points), reference scenario with the measured V1 PSF, matched filter. Bars: total (S_T) and first-order (S_1) indices; error bars: 95% bootstrap confidence intervals (SALib). Parameters ranked by S_T at 10^{-4}. Points not reaching the target within 1 s at survival ≥ 0.99 (31% at 10^{-4}, 4% at 10^{-3}) are censored at $\log_{10} t = 0$.

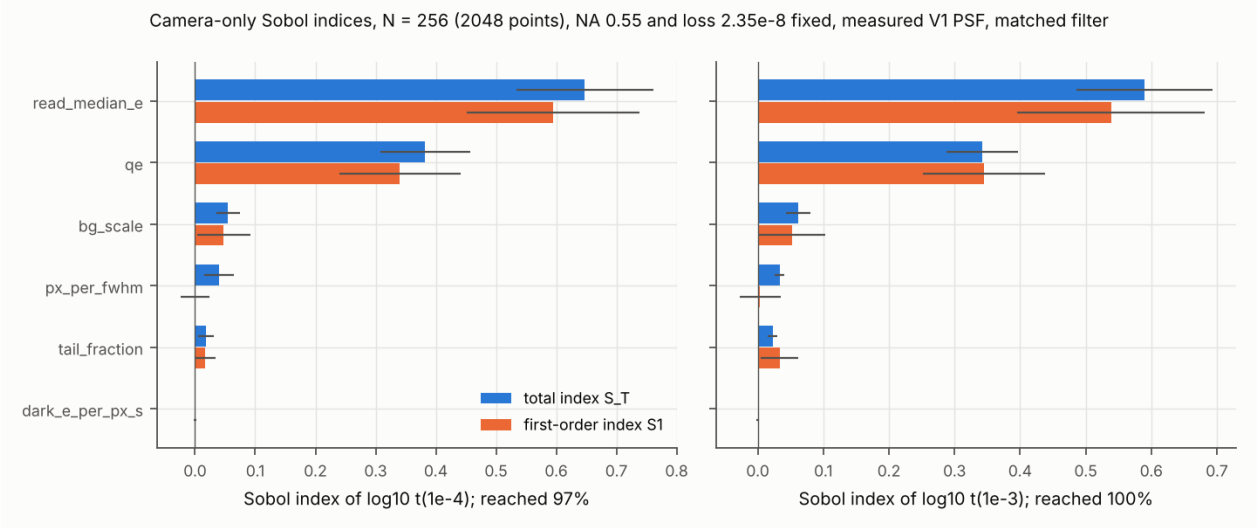


Figure 3: As Fig. 2 for the camera-only design: NA 0.55 and loss 2.35×10^{-8} per photon fixed, six camera-side inputs, $N = 256$ (2048 points), scan to 5 s (3% of points censored at 10^{-4}, none at 10^{-3}).

The response is additive (sum of $S_1 = 0.97$ at 10^{-3} and 1.00 at 10^{-4}). However, the ranking of read noise above QE reflects the sampled ranges: the read-noise range spans a factor of 40 and includes cameras that are clearly unsuitable, and the log-uniform background range has its median (0.55 of the reference) on the dark side, where read noise weighs more. For a specification, the local behaviour at a realistic operating point is the relevant quantity (Table 5).

Table 5: $t(10^{-3})$ / $t(10^{-4})$ in ms versus read-noise median at the reference background, 1.0 px per FWHM, no tail, dark current 0.01 e^-/px/s; same model as the Sobol designs.

QE	0.1 e^-	0.25 e^-	0.5 e^-	1.0 e^-	2.0 e^-
0.485	20.1 / 31.4	20.7 / 32.0	22.6 / 34.1	28.1 / 40.7	41.5 / 57.8
0.85	11.5 / 17.8	11.8 / 18.2	12.9 / 19.3	16.0 / 23.0	23.6 / 32.7

As can be seen from Table 5, the read-noise factor relative to 1.0 e^- is the same at both QE values to within 0.01 (for 10^{-4}: 0.77, 0.79, 0.84, 1, 1.42; for 10^{-3}: 0.72, 0.74, 0.80, 1, 1.48), and the QE dependence is 1/QE to within 1% (DERIVED from Table 5). Hence, at this operating point,

$$t(10^{-4}) \simeq 40.7\,\text{ms} \times \frac{0.485}{\text{QE}} \times f_4(\sigma_r), \tag{2}$$

where σ_r is the read-noise median and f_4 is the factor above. The same model reproduces the preset results (Prime BSI Express 41.3 ms; ORCA-Quest 2 PNR 39.3 ms against 40.2 ms for the preset, whose read-noise model includes a tail class).

7 Implications

Stage 1 specification (provisional; reference scenario, matched filter). Our choice of requirements is a compromise between imaging time and what a purchased back-illuminated CMOS sensor can deliver. First, QE at 852 nm sets the imaging time in proportion: each relative gain in QE shortens the time by the same factor. Second, a read-noise median of 0.5 e^- or below is within 10% of an ideal sensor, whereas 1.0 e^- costs about 30% and 2.0 e^- about 80%. Third, pixels with read noise far above the median must be rare: at a 1.0 e^- median, 2% of pixels at 5 times the median cost 42%; the acceptable fraction depends on the measured tail shape, which P4 must supply, and on whether such pixels can be masked. Fourth, cooling need only keep dark current below about 1 e^-/px/s, since the dark charge at 40 ms is then below 0.04 e^- per pixel (DERIVED); the Sobol designs show no effect of dark current up to 3 e^-/px/s. Fifth, the magnification should give 0.5 to 1.4 px per PSF FWHM (within 4% of the best in the camera-only design). The quantitative specification (minimum QE and maximum read noise for a given time budget) follows from Eq. (2) once the lab fixes the imaging-time budget; the generic sweep with the measured PSF (Sec. 9) will give the full maps.

Prime BSI Express operating point (provisional). CMS mode (1.0 e^-), magnification giving about 0.7 to 1.0 px per PSF FWHM (2.7 to 3.8 for the V1 PSF of 1.70 µm FWHM; to be rescaled with our own measured PSF), matched filter with per-site weights and thresholds: $t(10^{-3}) = 28.4$ ms, $t(10^{-4}) = 41.3$ ms, and $\varepsilon = 1.2 \times 10^{-4}$ at 40 ms in the reference scenario. Speed mode costs 19% (49.3 ms). A 2048×128 region of interest reads out in 1.46 ms (128 rows at 11.4 µs, USER datasheet line time); with continuous illumination the exposure must exceed the imaging time by this amount. The read-noise tail of the GSENSE2020BSI is not specified; a dark-frame stack from our camera is the most useful next input.

Programme level. At the operating point, raising QE from 0.485 to 0.85 shortens $t(10^{-4})$ by a factor of 1.77 (Table 5). Raising NA from 0.55 to 0.70 increases the collection fraction by a factor of 1.7 (0.083 to 0.143 for isotropic emission, DERIVED), a comparable gain from the optics. In our

opinion, the case for a custom thick-silicon sensor therefore rests on QE at 780 and 852 nm together with cost and supply, and the gain in detection speed alone (about 1.8 times over a good commercial back-illuminated sensor) should be weighed against an NA upgrade.

8 Limitations

(i) The V1 FP rate is under-predicted 14-fold; if real experiments carry such a floor, camera improvements below it are invisible. (ii) Read-noise tails are a two-class model (tail pixels at 3 times the median for presets with a stated rms, 5 times in the generic and Sobol designs); P4 will replace this with measured distributions. (iii) All production results are for one reference scenario: Cs at 852 nm, NA 0.55, the measured V1 PSF, a uniform background, no neighbour crosstalk, filling 0.5. (iv) The characteristic-function engine omits ADC rounding; for the ORCA-Quest 2 in PNR mode exact rounding lowers ε by 7% at the operating point (1 to 2% in time), so the PNR results are slightly conservative. (v) The Sobol results are at $N = 256$, with half-widths of 0.04 to 0.15 on the leading indices. (vi) Model B (Langevin) is used only to bound the effect of the loss-time profile; the production results use model A.

9 Remaining work

Table 6: Outstanding computations and deliverables.

item	size	why it is needed
generic CMOS sweep with the measured PSF	725 of 750 points left, at most 19 CPU-h	spec-to-fidelity maps in the (QE, read noise) plane consistent with Table 2; the existing maps use the diffraction-limited PSF
full 8-parameter Sobol, $N = 1024$	7680 points left, at most about 60 CPU-h	halves the confidence intervals; separates NA, loss and read noise at 10^{-4}
full camera-only Sobol, $N = 1024$	6144 points left, at most about 30 CPU-h	final ranking of camera parameters for the specification
Rb at 780 nm and the Cs/Rb compromise	not started	required by the brief; QE and PSF differ at 780 nm
neighbour crosstalk and array geometry	not started	required by the brief; matters when site spacing approaches the PSF width
Bayesian per-pixel classifier in production	implemented, validated, not yet in the sweeps	can exploit a measured read-noise map, the input that matters most after QE
EMCCD branch of the generic sweep	not started	EM gain, CIC and frame rate as free parameters
written Stage 1 specification and Prime BSI Express recipe	after the items above	the two deliverables of the brief

The three runs are prepared for the IIT Madras HPC (`runs/production/slurm_submit_measpsf.sh`, 8 tasks × 16 cores by default; partition and account [CONFIRM?]). Since the runs above, the time-to-target scan stops after the last first crossing; re-evaluated points give identical times at 4 to 20 times lower cost, so the sizes in Table 6 are upper bounds.

10 Computational cost

Table 7: CPU time used on the 4-core cloud instance.

study	points	CPU-h
presets, measured PSF	77	0.66
sensitivity, measured PSF	26	0.17
generic sweep, diffraction-limited PSF	750	7.3
8-parameter Sobol, $N = 256$	2560	20.3
camera-only Sobol, $N = 256$	2048	11.0

References

[1] Manetsch, Nomura, Bataille, Leung, Lv and Endres, "A tweezer array with 6100 highly coherent atomic qubits", Nature **647**, 60–67 (2025), doi:10.1038/s41586-025-09641-4; text read: arXiv:2403.12021v4.

[2] Picken, Legaie and Pritchard, "Single Atom Imaging with an sCMOS camera", Appl. Phys. Lett. **111** (2017), doi:10.1063/1.5003304; text read: arXiv:1709.00946v2.

[3] Basden, Haniff and Mackay, "Photon counting strategies with low light level CCDs", Mon. Not. R. Astron. Soc. **345**, 985–991 (2003), doi:10.1046/j.1365-8711.2003.07020.x; text read: arXiv:astro-ph/0307305v1.

[4] Steck, "Cesium D Line Data", revision 2.3.4 (2025), `https://steck.us/alkalidata/cesiumnumbers.pdf`.

[5] SALib, Sensitivity Analysis Library in Python, version 1.6.0, `https://github.com/SALib/SALib`.

A Provenance ledger (principal values)

value	location	type	pointer
FN 5.50×10^{-5}, FP 7.01×10^{-5}, fidelity 0.9999374(8), QE 0.44, 2.7%, 80 ms, NA 0.55	Sec. 2, 3	SOURCE	Ref. [1], arXiv:2403.12021v4 (main text and Methods)
readout error $< 8 \times 10^{-7}$, 40 ms, Zyla 5.5	Sec. 2	SOURCE	Ref. [2], arXiv:1709.00946v2
$F^2 = 1/g + 2/g^{1/r} - 2/g^{1+1/r}$	Sec. 2	SOURCE	Ref. [3], Eq. 5 (`sources/README.md`)
FN 4.91×10^{-5} to 5.73×10^{-5}, FP 4.85×10^{-6}, gains 0.381 / 0.368 e^-/ADU, $F^2 = 1.972$	Table 1	SIM	`runs/validation/`, `reports/p1_validation.md` (review before submission)
rate 72.3×10^3/s, η_{opt} 0.327, loss 2.35×10^{-8}, background 4.76 e^-/px/s	Sec. 3	DERIVED / DATA	`configs/production/reference_cs.yaml` pointers
PSF $1/e^2$ radius 1.44 µm, FWHM 1.70 µm; vectorial FWHM 0.82 µm	Sec. 3	DERIVED	`reference_cs.yaml`; `optics/array.py` (PSFSpec.fwhm)
survival ≥ 0.99 up to 5.9 s	Sec. 3	DERIVED	$0.01005/(2.35 \times 10^{-8} \times 72.3 \times 10^3/\text{s})$
QE at 852 nm per preset	Table 2	DATA / USER	`camsim_common/presets/*.yaml` (digitised vendor curves; USER-supplied PDFs)
preset times, magnifications	Table 2, Fig. 1	SIM	`runs/production/prod_presets/`, `prod_presets_difflim/` (review before submission)
sensitivity times	Table 3	SIM	`runs/production/prod_sensitivity/` (review before submission)
loss bound 5.6×10^{-7}	Table 3	DERIVED	Ref. [2]: retention > 99% at 450 photons/ms, 40 ms (`configs/production/sensitivity.yaml`)
8-parameter Sobol indices	Sec. 6, Fig. 2	SIM	`runs/production/prod_sobol/points/`, rows 0 to 2559 (review before submission)
camera-only Sobol indices	Table 4, Fig. 3	SIM	`runs/production/prod_sobol_camera/points/`, rows 0 to 2047 (review before submission)
operating-point times	Table 5	SIM	`p1_atomdet.sweep.evaluate.sobol_point` with `configs/production/sobol_camera.yaml` (review before submission)
read-noise factors, 1/QE scaling, Eq. (2)	Sec. 6	DERIVED	ratios of Table 5 entries
collection fraction 0.023, 0.083, 0.143, 0.200 at NA 0.3, 0.55, 0.7, 0.8	Sec. 6, 7	DERIVED	$(1 - \sqrt{1 - \text{NA}^2})/2$, isotropic emission (`optics/collection.py`)
line time 11.4 µs (CMS)	Sec. 7	USER	Prime BSI Express datasheet rev. 2024-08-20
CPU time	Table 7	DATA	per-point cost records in the run directories

B Open placeholders

The presets carry 116 open [VALUE?] entries (full list in `reports/p1_production_note.md`). Of these, 40 belong to the two validation presets and the model-B parameters and are discussed in `reports/p1_validation.md`; 76 belong to the eleven production presets. The production engine does not read 54 of the 76 (cosmic-ray parameters, PRNU, ADC offset, line time). The remaining 22 can affect the results above: (i) read-noise median or rms where the vendor states a single value (6 entries; sets the tail class of the preset); (ii) dark-current parameters (7; negligible by Sec. 6 unless dark current exceeds about 1 e^-/px/s); (iii) EMCCD register length and CIC split (4; the split is swept in Table 3); (iv) ADC parameters (5; ADC rounding is omitted by the production engine). In addition, the SLURM partition and account are [CONFIRM?].